

Institute of Science and Technology (IST)

B.Sc (Hons) in CSE, Part-I, 2nd Semester

Course name: Linear Algebra (CSE-510225)

Time: 3 hours

Full Marks: 80

1. a) Define homogeneous and non-homogeneous linear equation. $3+5=8$
Reduce the following system of linear equation into echelon form and solve it:

$$x + 2y + 3z = -1$$

$$2x - y - 2z = 5$$

$$3x + 3y + z = -2$$

$$4x + 3y + 4z = 3$$

$$-4x + 2y + 4z = -10$$

- b) Determine the values of λ such that the following system in unknowns x, y and z has

i) a unique solution, ii) no solution,

iii) more than one solution:

$$\lambda x + y + z = 1$$

$$x + \lambda y + z = 1$$

$$x + y + \lambda z = 1$$

- c) Prove that

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} = (a-b)(b-c)(a-c)(a-d)(b-d)(c-d)(a+b+c+d)$$

2. a) Define (i) Matrix, (ii) Transpose of a matrix, (iii) Identity matrix, (iv) Orthogonal matrix.

- b) Prove that every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew symmetric matrix.

c) If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ then prove that $A^3 - 4A^2 - A + 4I = 0$ and hence find A^{-1} .

3. a) Define norm of a vector in $\mathbb{R}^n$. Prove that $|u+v| \leq |u| + |v|$ 6
 b) State and prove Cauchy Schwarz inequality. 8
 c) Define rank of a matrix. Find the rank of the matrix 1+5=6

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 0 & 1 \\ 3 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 \end{bmatrix}$$

4. a) Define Linear transformation Show that the following transformation defines a linear operator of $\mathbb{R}^3$ 2+5=7
 $T(x, y, z) = (x+y, -x-y, z)$

- b) Define rank and nullity of linear transformation. If $T: U \rightarrow V$ is a linear transformation then show that rank of T + nullity of $T = \dim U$. 2+6=8

- c) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x+2y-z, y+z, x+y-2z)$. Find a basis and the dimension of (i) $\text{Im}T$ of T (ii) $\text{Ker}T$ of T . 5

5. a) Define matrix representation of a linear operator. Let $\{u_1, u_2, \dots, u_m\}$ be a basis of U and T be any linear operator of U , then for any vector $w \in U$, show that $[T]_u[W]_u = [T(W)]_u$. 3+7=10

- b) The mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by $T(x, y, z) = (2y+z, x-4y, 3x)$. Show that T is a linear transformation and find the matrix of T in the basis $\{u_1=(1,1,1), u_2=(1,1,0), u_3=(1,0,0)\}$. 4+6=10

6. a) Define eigenvalue and eigenvector. State and prove Cayley-Hamilton theorem. 2+6=8

- b) Find the characteristic equation of the matrix 12

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & 1 \end{bmatrix} \text{ and verify Cayley-Hamilton theorem for it.}$$